



[Sign up for this introductory AI course to gain practical, industry-ready AI skills you'll need to increase productivity, innovate, create, and succeed. These skills include understanding how modern AI systems work, learning what is AI and what is not, using leading AI tools for research, creating study guides for deeper subject matter understanding, evaluating AI's strengths and limitations, and applying ethical and faith-informed principles to real-world AI challenges.]

How to get the most out of [A Catholic Introduction to AI, Part 2](#):

- Read the assignments before class meetings.
- Have a notebook ready and available for class notes during each live or recorded session.
- Do the assignments, quizzes, and any extra work assigned for that week.

Once the course is completed to the parent's and instructor's satisfaction, there is a Certificate of Completion at the end to be filled in for your records.

Total Classes: 13 Weeks

Class Day and Dates: Tuesdays, January 5 to April 13, 2027. (No class Feb. 9 for spring break, and Mar. 23 for Holy Week)

Starting Time: 4:00 PM Eastern (3:00 Central; 2:00 Mountain; 1:00 Pacific)

Duration: 60 minutes

Suggested Grade Level: High School

Suggested Credit: 2 full semesters Computer Science

Instructor: Mr. Daniel Goddu, BS-CS

Textbook: None; all materials provided with the course with no extra cost

Prerequisites: One of the following or equivalent:

- Catholic Introduction to Artificial Intelligence, Part 1

Course Requirements:

This course uses the following AI-based tools:

- **NotebookLM** is an AI research and study tool developed by Google that helps users understand and work with their own documents. Instead of answering questions from the general internet, it analyzes sources you upload—such as PDFs, Google Docs, or notes—and generates summaries, explanations, study guides, and even audio-style discussions based only on those materials. This makes it especially useful for research, learning, and



organizing information because the AI stays grounded in the sources you provide. **A Google account is required.**

- **Jupyter Notebook** is an open-source interactive computing environment where users can write code, display results, add explanations, and create visualizations all in one document. It is widely used in data science, machine learning, and AI because it allows programmers to run code step-by-step while documenting their work. This makes it especially useful for learning, experimentation, and sharing reproducible research.

Jupyter Notebooks are best created and executed using either **Visual Studio Code** (software download required, no account required) or **Google Colab** (Google account required). Refer to the description of these platforms below (choose either VSC or Colab).

- **Visual Studio Code (VSC):** VSC is an integrated development environment (IDE) that aids in the writing, debugging, and execution of a large range of computer programs. It also can handle Jupyter Notebooks. Students with computer programming experience and who are familiar with VSC while find using VCS for Jupyter Notebooks quite easy. VSC software is free but it must be installed on the student's computer or laptop.
- **Google Colab:** (short for *Google Colaboratory*) is a free cloud-based environment from Google that lets users write and run Python code in a web browser using Jupyter Notebooks. Because it runs in the cloud and can provide access to GPUs, students can build and test AI models without installing software on their own computers. **A Google account is required.**

Access to the following open-source Large Language Models (LLMs):

- **Ollama (<https://ollama.com/>):** Ollama (OLLaMA) is an **open-source**, large-scale language model developed by Alibaba Cloud. It can generate natural-sounding text responses based on the prompt provided, making it capable of understanding and engaging with complex topics across multiple domains. **An Ollama free-usage account is required.**

Access to **one** of the following cloud-based proprietary AI models:

- **ChatGPT (<https://chatgpt.com/>):** an AI assistant that explains ideas, answers questions, researches information, and helps with writing, learning, planning, and problem-solving across many topics (from Chat GPT). **No account registration required.**
- **Google Gemini (<https://gemini.google.com/app>):** I am Gemini 3 Flash, an adaptive AI collaborator. I'm here to help you create text, images, music, and video with clarity and wit (from Gemini). **A Google account is required.**
- **Claude (<https://claude.ai/login>):** an AI assistant made by Anthropic. I can answer questions, write, analyze, code, and help with a wide range of tasks (from Claude). **Account registration is required.**



Course Description:

Having now an introduction to the foundations of artificial intelligence and some hands on experience with building and working with modern AI tools, Part Two focuses on how large language models operate in practice by running open-source models locally using tools such as Ollama and interacting with them through interfaces like Open WebUI. They explore prompt engineering, system prompts, tools, and functions, and experiment with AI-powered automation. The semester concludes with projects in which students design simple AI assistants or agents, giving them practical experience applying AI responsibly and creatively.

Students will engage in **interactive discussions** and **hands-on AI experiments**. The class will encourage **critical thinking about AI's benefits, risks, and ethics**. While students will not be required to write any computer programs, a course in computer programming is required as a prerequisite. A recorded tutorial on Python and SNAP! Are part of this course to bring the student up to a level of the required proficiency to be successful for the course. Programming examples are provided. The student will learn and be evaluated on what AI objective the code achieves.

This course integrates faith and technology to explore faith, ethics, and AI advancements together. The Catholic Church views science to explore and comprehend the natural world created by God. Technology is encouraged when it respects ethical principles and upholds the dignity of human life. Catholics recognize the benefits of technological advancements for improving human life and society. Moral discernment is crucial to guide technology to serve humanity ethically and avoid harm or injustice.

This course will not teach students how to exploit AI concepts that violate academic integrity. Its main goal is to introduce the student to AI concepts and the technology that makes it happen. Over the course of the topics, students will understand the so-called *truth* that AI provides and the truth that it doesn't provide. And it will cut through the fantasy and hype so often attributed to AI.

Semester 2: Applied AI & Building with AI

Class 14: Introduction to LLMs

Students revisit the inner workings of large language models with a deeper dive into tokenization, context windows, and prompt-response pipelines. They explore how LLMs generate text by predicting the next word and how context length limits coherence. Comparisons between proprietary and open-source models prepare students for self-hosting.

Class 15: Using Open-Source LLMs with Ollama

Students are introduced to Ollama, a tool that simplifies running LLMs locally. They learn how to install Ollama, load models like Mistral or Llama 3, and query them from the command line. Students gain insight into the hardware requirements, performance trade-offs, and offline benefits of local LLMs.



Class 16: Open WebUI Frontend – Basics

In this class, students install and launch Open WebUI as a user-friendly interface to interact with local models. They explore its chat layout, prompt history, and simple configuration settings. This class emphasizes building user-facing tools with privacy and customization in mind.

Class 17: System Prompts, Roles, and Memory

Students explore how LLM behavior can be shaped using system prompts and defined roles. They experiment with prompt engineering to control tone, persona, and output length. The concept of memory (temporary vs. persistent context) is introduced and tested.

Class 18: Tools & Functions with LLMs

This session focuses on expanding what LLMs can do using tools and function-calling. Students learn how Open WebUI enables models to access external tools like calculators, file loaders, and custom plugins. They create simple tools and test calling them from within the chat.

Class 19: Prompt Engineering

Students develop advanced prompting skills. They learn how to craft instructions with few-shot examples, control outputs with formatting techniques, and guide reasoning using step-by-step prompts. Side-by-side comparisons help students evaluate effectiveness.

Class 20: AI Automation

In this class, students will create workflows that integrate AI models with common tools like email, forms, or Google Docs. They explore scenarios like generating meeting summaries, automating feedback forms, or batch-creating documents with LLM input.

Class 21: AI Agents – What Are They?

Students are introduced to the concept of autonomous agents—LLMs that act on goals using tools and memory. They learn about agent architecture (planner, memory, executor), and evaluate tools like LangChain, AutoGen, and Agentops. Use cases include research assistants and data scrapers.

Class 22: Build a Simple Agent (Final Project Work Session, Part 1)

Students design and deploy a basic agent using Ollama API-based functions. Agents might be



tasked with designing a “rule of life” (prayer), Bible study companion, or a homework planning assistant.

Class 23: Final Project Work Session, Part 2

This class is devoted to building and refining student projects. Students work individually on an applied AI solutions choosing one of the above projects.

Class 24: Final Project Work Session, Part 3

Students focus on debugging prompt issues, interface errors, or tool integration problems. Students document challenges and solutions as part of their learning process.

Class 25: Final Presentations I

Half of the class presents their final projects. Each student shares their problem statement, how AI was used, what tools were integrated, and a short live or recorded demo. Peer feedback is encouraged.

Class 26: Final Presentations II & Course Reflection

The rest of the class presents, followed by a course-wide reflection. Students discuss what surprised them, what changed their minds, and how they plan to use AI responsibly going forward.

Assessment Overview

- **Short essays – 30%**
- **Quizzes – 10%**
- **Labs – 20%**
- **Final Project – 40%**

Bibliography

[1] Russell, Stuart, and Peter Norvig. Artificial Intelligence: A Modern Approach. Pearson, 2020.

[2] Muelier, John Paul, Luca Massaron, and Stephanie Diamond. Artificial Intelligence for Dummies. John Wiley & Sons, Inc. 2025.



[3] Poole, David L, and Alan K Mackworth. Artificial Intelligence 3E: Foundations of Computational Agents. Cambridge University Press, 2023. <https://artint.info/>

[4] Bulloch, Seth. Artificial Intelligence: Teaching Artificial Intelligence to High School Students. Kindle Ed.